

Security risk assessment report

Part 1: Select up to three hardening tools and methods to implement

The firewalls currently have no rules in place to filter traffic coming in and out of the network. We should configure firewall rules to allow required traffic and block unauthorized traffic entering and leaving the network.

The organization's employees share passwords and the database admin account uses a default password. Network access privileges should be implemented to limit network access based on roles, groups, people, IP addresses, MAC addresses etc.

There is no second line of defence in the event that a threat actor gains hold of an employee's password. Multifactor authentication (MFA) should be implemented to improve protection so that if a threat actor gains access to a password, they would still need to authenticate the login credentials using a second form of identification such as a one-time password (OTP) to gain access.

Part 2: Explain your recommendations

By regularly updating the firewall, this allows us to stay ahead of any potential threats and mitigate the risk of a future attack. Whilst a firewall does not prevent all potential threats, it does work as an effective security layer against network traffic entering the network. The rules filter traffic continuously and should be reviewed periodically and when network requirements change.

By applying multifactor authentication (MFA), a user will have to use two different authentication factors such as a password and a one-time passcode (OTP) in order to gain access to the system. This makes it harder for an attacker because even if they correctly enter the password, they will still need to approve the login from another device or service.